and so, for central forces,

$$\sum_{i=1}^{\mathrm{N}} m_i \ddot{\boldsymbol{q}}_i = \mathbf{0},$$

which expresses the constancy of the momentum. Moreover, viewing the $\boldsymbol{q}_i$ as vectors in $\mathbb{R}^3$ (with 0 as their third component),

$$\boldsymbol{q}_i \times \frac{\partial}{\partial \boldsymbol{q}_i} U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j) = -\boldsymbol{q}_j \times \frac{\partial}{\partial \boldsymbol{q}_j} U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j),$$

which implies

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{\mathrm{N}} \boldsymbol{q}_i \times (m_i \dot{\boldsymbol{q}}_i) = \mathbf{0},$$

and tells us that the total *angular momentum* is also conserved. This means that the system will translate and rotate at a constant rate in time.[6] By a judicious choice of initial conditions, the center of mass can be kept at the origin and the net rotation removed. This corresponds to a symmetric configuration which is termed the *isosceles* three body problem (i.e., at all times the atoms are the vertices of an isosceles triangle). The original system had six positional degrees of freedom, but four have now been removed by imposing the momentum constraints. Therefore we would expect a model with just two degrees of freedom. While the conservation of angular and total momentum allow us to remove some of the degrees of freedom of the problem, reducing it to a smaller system, this problem is not integrable.

Let us introduce new coordinates in the Lennard-Jones trimer as illustrated in Fig. 1.21, so that the center of mass is fixed at the origin: that is,

$$\boldsymbol{q}_1 = \begin{bmatrix} x \\ -y/3 \\ 0 \end{bmatrix}, \quad \boldsymbol{q}_2 = \begin{bmatrix} -x \\ -y/3 \\ 0 \end{bmatrix}, \quad \boldsymbol{q}_3 = \begin{bmatrix} 0 \\ 2y/3 \\ 0 \end{bmatrix},$$

reducing the energy to

$$E = \dot{x}^2 + \frac{\dot{y}^2}{3} + 2\hat{\varphi}_{\mathrm{LJ}}(\sqrt{x^2 + y^2}) + \hat{\varphi}_{\mathrm{LJ}}(2x),$$

which describes the vibrational motion. Over time the positions will vary within some region in the xy plane, whose boundary is dependent on the energy E of the initial condition. What is this boundary? Since the energy is conserved, and the

[6] Note that in periodic boundary conditions the linear momentum would also be preserved, but the angular momentum is not.